

THEORY AND METHODS

ON THE NORM OF THE ALIAS MATRIX OF A FRACTIONAL FACTORIAL DESIGN UNDER THE MOST GENERAL PARTITIONED LINEAR MODEL

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Summary: This paper presents results on the norm of the alias matrix A_D of a fractional factorial design D of the general $s_1^{k_1} \times s_2^{k_2} \times \dots \times s_t^{k_t}$ factorial under the general partitioning of the parametric vector $\beta' = (\beta'_1 | \beta'_2 | \beta'_3)$, where β_1 is a $t_1 \times 1$ vector of effects to be estimated, β_2 is a $t_2 \times 1$ vector of effects with doubtful negligibility and not of interest for estimation and β_3 is a $t_3 \times 1$ vector of effects assumed to be negligible, where $1 < t_1 \leq s$, $0 \leq t_2 \leq s-1$, $0 \leq t_3 = s - t_1 - t_2 \leq s-1$, and $s = \pi_{i=1}^t s_i^{k_i}$. Explicitly $A_D = M_D^{-1} [X'_{D1} X_{D3} - X'_{D1} X_{D2} (X'_{D2} X_{D2})^{-1} X'_{D2} X_{D3}]$ in $E[\hat{\beta}_1] = \beta_1 + A_D \beta_3$ if the assumption that β_3 is equal to zero is in doubt; here the information matrix is $M_D = [X'_{D1} X_{D1} - X'_{D1} X_{D2} (X'_{D2} X_{D2})^{-1} X'_{D2} X_{D1}]$, where X_{D1} , X_{D2} , X_{D3} are the design matrices relative to β_1 , β_2 and

β_3 . The norm $\|A_D\| = [\text{trace}(A_D' A_D)]^{\frac{1}{2}}$, is used as an optimality criterion in selecting a design since it measures alias goodness of a design in estimating β_1 . It is shown that $\|A_D\|$ can be expressed in terms of the information matrix M_D for β_1 and it does not involve the design matrix X_{D3} . For special classes of designs it is shown that minimizing $\|A_D\|$ is equivalent to minimizing $\text{trace}(M_D^{-1})$. The results of Shirakura (1976) correspond to the special case $t_2 = 0$.

1. INTRODUCTION

Consider the general $s_1^{k_1} \times s_2^{k_2} \times \dots \times s_t^{k_t}$ factorial, $s_i \geq 2$, where each of the k_i factors has s_i levels from the set $S_i = \{0, 1, 2, \dots, s_i - 1\}$. Let $S_i^{k_i} = S_i \times S_i \times \dots \times S_i$, that is the Cartesian product of S_i with itself k_i times. Further, let the set of treatment combinations be represented by the Cartesian product $S = S_1^{k_1} \times S_2^{k_2} \times \dots \times S_t^{k_t}$. Note that S is a minimal complete factorial design (or a single complete replicate) with $s = |S| = \pi_{i=1}^t s_i^{k_i}$ treatment combinations.

Let Y_S be the observational vector for the minimal complete factorial design S and let β be the $s \times 1$ vector of parameters reflecting the factorial effects. Assume the linear model for Y_S , that is

$$(1.1) \quad Y_S = X_S \beta + \epsilon_S \\ E(\epsilon_S) = 0, \quad \text{Cov}(\epsilon_S) = \sigma^2 I_S,$$

where X_S is the $s \times s$ design matrix of rank s , and I_s is the identity matrix of order s . The design matrix X_S can be regarded as an orthogonal matrix.

Now let D be any factorial design, that is a collection of n treatment combinations from S , and let β be partitioned into three parts, that is

$$(1.2) \quad \beta' = (\beta'_1 \mid \beta'_2 \mid \beta'_3),$$

where β_1 is a $t_1 \times 1$ vector of factorial effects to be estimated, β_2 is a $t_2 \times 1$ vector of factorial effects with doubtful negligibility and not of interest for estimation (thus β_2 is a vector of nuisance parameters), and β_3 is a $t_3 \times 1$ vector of factorial effects assumed to be negligible, $1 < t_1 \leq s$, $0 \leq t_2 \leq s-1$, $0 \leq t_3 = s - t_1 - t_2 \leq s-1$. The model for the observational vector Y_D , corresponding to the fractional factorial design D , is implied by (1.1) and (1.2), that is

$$(1.3) \quad Y_D = (X_{D1} \mid X_{D2}) \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} + \epsilon_D$$

$$E(\epsilon_D) = 0, \quad \text{Cov}(\epsilon_D) = \sigma^2 I_n,$$

where the matrices X_{D1} and X_{D2} are simply read off from X_S in (1.1) taking repetitions into account. Using the least squares procedure it is easy to show that the information matrix of the fractional factorial design D relative to β_1 is given by

$$(1.4) \quad M_D = [X'_{D1}X_{D1} + X'_{D1}X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2}X_{D1}].$$

We will assume that the rank of the information matrix is equal to t_1 , so that the least squares estimate of β_1 will be equal to

$$(1.5) \quad \hat{\beta}_1 = M_D^{-1} [X'_{D1} - X'_{D1}X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2}] Y_D,$$

with covariance matrix of $\hat{\beta}_1$ equal to

$$(1.6) \quad \text{Cov}(\hat{\beta}_1) = [X'_{D1}X_{D1} - X'_{D1}X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2}X_{D1}]^{-1} \sigma^2$$

$$= M_D^{-1} \sigma^2.$$

In the above H^- denotes a generalized inverse of the matrix H .

Now, suppose that the assumption that β_3 is zero is false, so that the true model for a fractional factorial design D is given by

$$(1.7) \quad Y_D = X_{D1}\beta_1 + X_{D2}\beta_2 + X_{D3}\beta_3 + \epsilon_D, \quad \text{and}$$

$$E(\epsilon_D) = 0, \quad \text{Cov}(\epsilon_D) = \sigma^2 I_n.$$

Under this model $\hat{\beta}_1$ of (1.5) is no longer unbiased and it can be shown that

$$(1.8) \quad E(\hat{\beta}_1) = \beta_1 + A_D \beta_3$$

where

$$(1.9) \quad A_D = M_D^{-1} [X'_{D1} X_{D3} - X'_{D1} X_{D2} (X'_{D2} X_{D2})^{-1} X'_{D2} X_{D3}].$$

The matrix A_D is called the *alias* or *contamination matrix* of the design D relative to β_1 , β_2 and β_3 . Since $\hat{\beta}_1$ is a biased estimator of β_1 the appropriate quantity now is the mean square error of $\hat{\beta}_1$, which is equal to

$$(1.10) \quad MSE(\hat{\beta}_1) = M_D^{-1} \sigma^2 + A_D \beta_3 \beta_3' A_D'.$$

It is clear that selection of an optimal design will require a criterion which will take both parts in (1.10) into account. We will not go into the optimal design selection problem as indicated, but the reader can find a detailed discussion in the recent paper by Raktøe and Pesotan (1993).

If bias is of major concern in selecting an optimal design from a well-defined class of competing designs with the assertion that the experimenter is ignorant about the magnitudes of the effects in β_3 , then an optimality criterion should be based on $A_D A_D'$, as is typically done in the unbiased case by basing a criterion on M_D^{-1} , since σ^2 is unknown. Raktøe and Pesotan (1993) have suggested the use of $trace(A_D A_D')$ as an optimality criterion for finding an *alias-optimal* design. They pointed out that:

- (i) $trace(A_D A_D') = trace(A_D' A_D) = (\|A_D\|)^2 =$ the sum of the eigenvalues of $A_D' A_D$, where the Euclidean norm $\|A_D\|$ is equal to

$$(1.11) \quad \|A_D\| = \left[\sum_{i=1}^t \sum_{j=1}^t a_{ij}^2(D) \right]^{\frac{1}{2}}, \text{ and}$$

- (ii) minimizing $trace(A_D' A_D)$ is equivalent to minimizing the norm of A_D , viz. $\|A_D\|$.

In the next section we establish a general theorem on $\|A_D\|$ which leads to several interesting corollaries.

2. THE NORM OF THE ALIAS MATRIX

Let $Z'_D = X'_{D1}(I - X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2})$, so that $Z'_D Z_D = M_D$ and let z_{ij} be the row of Z_D , corresponding to the i -th treatment combination in S , j being the ordering of the distinct treatment combination in D . We are now ready to prove the following theorem.

THEOREM. Let D be a design with n treatment combinations, which has exactly the i_j -th treatment combination in S with multiplicity $r_{i_j} \geq 1$ for each $j = 1, \dots, m$, (that is $n = \sum_{j=1}^m r_{i_j}$), then the norm $|A_D|$ can be expressed as

$$|A_D|^2 = \text{trace}(M_D^{-1}) - t_1 + \sum_{j=1}^m r_{i_j}(r_{i_j} - 1)z'_{i_j} M_D^{-2} z_{i_j}$$

PROOF. From the model (1.7) we have

$$\begin{aligned}(X_{D1} | X_{D2} | X_{D3})(X_{D1} | X_{D2} | X_{D3})' &= X_{D1}X'_{D1} + X_{D2}X'_{D2} + X_{D3}X'_{D3} \\ &= \text{diag}(J_{r_{i_1}}, J_{r_{i_2}}, \dots, J_{r_{i_m}}) \\ &= T_D, \text{ say,}\end{aligned}$$

where J_r denotes the square matrix of order r with all elements equal to one. The alias matrix can be expressed as $A_D = (Z'_D Z_D)^{-1} Z'_D X_{D3}$. Then

$$\begin{aligned}A_D A'_D &= (Z'_D Z_D)^{-1} Z'_D X_{D3} X'_{D3} Z_D (Z'_D Z_D)^{-1} \\ &= (Z'_D Z_D)^{-1} Z'_D (T_D - X_{D1}X'_{D1} - X_{D2}X'_{D2}) Z_D (Z'_D Z_D)^{-1} \\ &= (Z'_D Z_D)^{-1} Z'_D T_D Z_D (Z'_D Z_D)^{-1} - I_{t_1} \\ &\quad \text{since } (Z'_D Z_D)^{-1} Z'_D X_{D1} = (Z'_D Z_D)^{-1} Z'_D Z_D = I_{t_1}, \\ &\quad \text{and } Z'_D X_{D2} = X'_{D1} (I - X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2}) X_{D2}\end{aligned}$$

$$\begin{aligned}
&= X'_{D1}(X_{D2} - X_{D2}(X'_{D2}X_{D2})^{-1}X'_{D2}X_{D2}) = 0 \\
&= (Z'_D Z_D)^{-1} + (Z'_D Z_D)^{-1} Z'_D T_D Z_D (Z'_D Z_D)^{-1} - (Z'_D Z_D)^{-1} - I_{t_1} \\
&= (Z'_D Z_D)^{-1} + (Z'_D Z_D)^{-1} Z'_D (T_D - I) Z_D (Z'_D Z_D)^{-1} - I_{t_1} \\
&= M_D^{-1} + M_D^{-1} Z'_D (T_D - I) Z_D M_D^{-1} - I_{t_1} \\
&= M_D^{-1} + \sum_{j=1}^m r_{i_j} (r_{i_j} - 1) M_D^{-1} z_{i_j} z'_{i_j} M_D^{-1} - I_{t_1}
\end{aligned}$$

$$\text{where } Z'_D T_D Z_D = \sum_{j=1}^m r_{i_j}^2 z_{i_j} z'_{i_j}, \quad Z'_D Z_D = \sum_{j=1}^m r_{i_j} z_{i_j} z'_{i_j}.$$

Note that this theorem shows that the norm $|A_D|$ does not depend on the matrix X_{D3} corresponding to the vector β_3 of assumed negligible effects.

The following corollaries are direct consequences of the above theorem.

COROLLARY 1. For the design D of theorem 1

$$|A_D|^2 \geq tr(M_D^{-1}) - t_1,$$

where the equality will hold if and only if $r_{i_1} = r_{i_2} = \dots = r_{i_m} = 1$.

Observe that this corollary makes the search for an alias-optimal design in a class of fractional factorial designs, each consisting of m distinct treatment combinations, extremely simple.

COROLLARY 2. For the design D of theorem 1, suppose $r_{i_1} = r_{i_2} = \dots = r_{i_m} = 1$. If the trace of M_D^{-1} is minimized, then $|A_D|$ is minimized.

From this corollary it follows that if an average variance optimal design (or a-optimal design) is found in a class of designs, each consisting of m distinct treatment combinations, then that design is also alias-optimal.

Before proceeding further let us go back to the general partitioning of

(1.2) of the parametric vector β . Explicitly we have the following four cases:

$$(1.12) \quad \begin{aligned} (i) \quad & t_1 = s, \quad t_2 = t_3 = 0 \\ (ii) \quad & t_2 = 0, \quad t_3 \neq 0 \\ (iii) \quad & t_2 \neq 0, \quad t_3 \neq 0, \quad \text{and} \\ (iv) \quad & t_2 \neq 0, \quad t_3 = 0. \end{aligned}$$

The designs of odd and even resolutions (see Raktøe, Hedayat and Federer (1981)) belong to cases (ii) and (iii) respectively. In case (iii) the alias matrix simplifies to

$$(1.13) \quad A_D = M_D^{-1} X'_{D1} X_{D3}.$$

Note that cases (i) and (iv) are not of interest with respect to bias, since $t_3 = 0$. Also cases (i) and (ii) do not contain nuisance parameters. Thus case (iii) is the most general case in the context of both nuisance parameters and bias. Hedayat, Raktøe and Federer (1974) considered case (ii) in proposing the norm $|A_D|$, for A_D in (1.13), as an optimality criterion for selecting an alias-optimal design. Raktøe and Pesotan (1993) have dealt with case (iii) and have proposed the use of the norm $|A_D|$ in (1.11) as an optimality criterion. In either case a design D is said to be alias-optimal in a well-defined class of competing designs if the corresponding $|A_D|$ is minimal.

COROLLARY 3. For the design D of the theorem with $t_2 = 0$, we obtain all the results of Shirakura (1976) as a special case of the theorem.

3. EXAMPLE

Consider the $s_1^k = 2^4$ factorial with the minimal complete factorial design consisting of $s = 2^4 = 16$ treatment combinations arising from four factors each at two levels, that is

$$S = \{(0000), (1000), (0100), (0010), (0001), (1100), (1010), (1001), (0110), (0101), (0011), (1110), (1101), (1011), (0111), (1111)\}.$$

We associate with Y_S the usual orthogonal design matrix

$$X = 2^{-\frac{1}{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \otimes 2^{-\frac{1}{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \otimes 2^{-\frac{1}{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \otimes 2^{-\frac{1}{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix},$$

where $\otimes$ is the Kronecker product.

Next consider a resolution IV design problem with $m = 11$ distinct treatment combinations. In such a design setting the partitioned parametric vector $\beta' = (\beta'_1 \mid \beta'_2 \mid \beta'_3)$ consists of

$$\beta'_1 = (\alpha \ \beta \ \gamma \ \delta) = \text{vector of main effects,}$$

$$\begin{aligned} \beta'_2 &= (\mu \ \alpha\beta \ \alpha\gamma \ \alpha\delta \ \beta\gamma \ \beta\delta \ \gamma\delta) \\ &= \text{vector of the mean and two-factor interaction effects, and} \end{aligned}$$

$$\begin{aligned} \beta'_3 &= (\alpha\beta\gamma \ \alpha\beta\delta \ \alpha\gamma\delta \ \beta\gamma\delta \ \alpha\beta\gamma\delta) \\ &= \text{vector of three and four-factor interaction effects.} \end{aligned}$$

Note that in this example $t_1 = 4$, $t_2 = 7$ and $t_3 = 5$. Let us compare the following designs using the norm $|A_D|$:

$$D^* = \{(0000), (0001), (0011), (0100), (0101), (0111), (1001), (1010), (1011), (1100), (1101)\}, \text{ and}$$

$$D^{**} = \{(0001), (0010), (0011), (0100), (0111), (1000), (1010), (1011), (1101), (1110), (1111)\}.$$

Using the equality from corollary 1, we have

$$|A_{D^*}|^2 = 24 \text{ and } |A_{D^{**}}|^2 = 4,$$

so that D^{**} is alias superior to D^* .

In this example we did not consider the problem of finding an alias-optimal design, that is if Δ_{11} denotes the set of all designs consisting of 11 distinct treatment combinations from among $2^4 = 16$ treatment combinations, then it is an interesting problem to find an alias-optimal design.

4. DISCUSSION

In the case where the fractional factorial design consists of m distinct treatment combinations from among $s = \pi_{i=1}^t s_i^{k_i}$ treatment combinations, there will be a well-defined set Δ_m of competing designs with $|\Delta_m| = \frac{s!}{(m!)(s-m)!}$. Here m is predetermined and must satisfy certain necessary conditions depending on the partitioning of β . The equality in corollary 1 involves a functional on the spectrum of the information matrix M_D . This results in a partitioning of Δ_m into equivalence classes, where all designs in an equivalence class have the same value of the functional on the spectrum of M_D . The general problem of determining the number of equivalence classes and the cardinality of such classes remains unsolved as pointed out by Raktoe and Pesotan (1993). Apart from this problem, these authors also point out other unsolved related problems.

5. ACKNOWLEDGEMENTS

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6. REFERENCES

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